

Advanced nanostructured surfaces for the control of marine biofouling: the AMBIO project

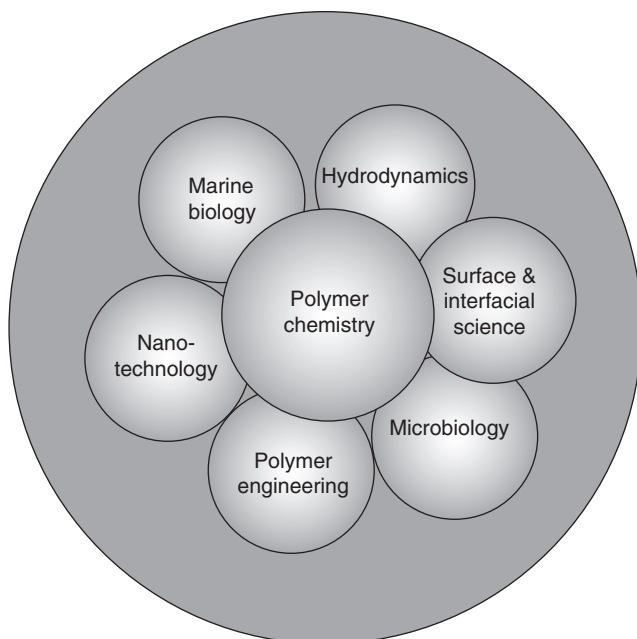
J A CALLOW and M E CALLOW,
University of Birmingham, UK

Abstract: The AMBIO project ('Advanced Nanostructured Surfaces for the Control of Biofouling') is an Integrated Project funded by the European Commission under its 6th Framework Programme. The 5-year project, which started in 2005, is devoted to the knowledge-based development of antifouling coatings that function through their nanoscale properties, without the use of biocides or materials that are released into the environment. The principal knowledge-driven strategy of the project is to achieve structure/property/performance correlations for a wide range of nanostructured coating designs. State-of-the-art analytical methods are used to characterise the coatings in aqueous 'seawater-like' environments. A wide range of laboratory-based settlement and adhesion assays are used to explore the antifouling potential of coatings, selected examples of which will enter comprehensive field trials for a range of end-use applications. The project involves 31 partner organisations (industries, universities and research institutes) linking the scientific expertise of polymer chemists and engineers, surface scientists, and marine biologists with the experience and commercial perspectives of coating technologists and raw materials manufacturers, together with a wide range of end-users. The project has a strong stakeholder orientation involving a range of end-user industries and suppliers of raw materials and technologies to ensure relevance and take-up of the results. This chapter outlines scientific progress towards the project goals in the first three years.

Key words: nanotechnology, nanocomposites, adhesion, fouling-release coatings, biofouling.

24.1 Introduction

The AMBIO project ('Advanced Nanostructured Surfaces for the Control of Biofouling') is an Integrated Project funded by the European Commission under its 6th Framework Programme. The 5-year project, which started in 2005, is devoted to the knowledge-based development of antifouling coatings that function through their nanoscale properties, without the use of biocides or materials that are released into the environment. The project

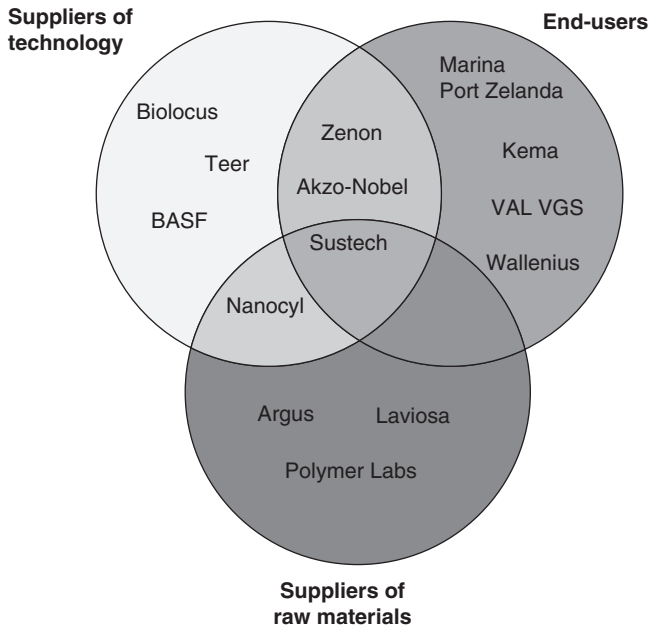


24.1 The interdisciplinary scientific base of AMBIO.

has a large critical mass and scale of resources: 31 partner organisations (industries, universities and research institutes) are linked as a legal consortium. The total value of the project is approx. €18 M, of which €12 M is provided by the EC. The project couples the scientific expertise of polymer chemists and engineers, surface scientists, hydrodynamicists, and marine biologists (Fig. 24.1) with the experience and commercial perspectives of coating technologists and raw materials manufacturers, together with a wide range of end-users (Fig. 24.2). The project has a strong stakeholder orientation involving a range of end-user industries and suppliers of raw materials and technologies to ensure relevance and take-up of the results.

24.1.1 Alternatives to biocides

Biocide-containing paints are the mainstay of the marine antifouling coatings industry and providing they meet relevant international legislation (e.g., in Europe the active ingredients have to comply with the Biocidal Products Directive EC 98/8/CE; Chapter 10), their use will continue for the foreseeable future. However, because the use of biocides is highly regulated, it is expensive to add new compounds to the list of allowed substances, which in turn means that there is restricted scope for innovation. The fol-



24.2 The company and supply-chain base of AMBIO.

lowing excerpt from the Biocidal Products Directive (BPD) is of particular relevance, a 'biocide' being defined as:

'Active substances and preparations containing one or more active substances, put up in the form in which they are supplied to the user, intended to destroy, render harmless, prevent the action of, or otherwise exert a controlling effect on any harmful organism by chemical or biological means.'

Alternatives to biocide-based technologies are therefore urgently sought. The implication of the above wording of the BPD is that it does not cover coatings that rely on purely physical means to create a surface with properties that reduce biofouling. Therefore, the ideal 'green' technology should be biocide-free, should not release any toxic ingredients into the environment and its effectiveness should be based on the physical properties of the coating. It is in the latter context that the 'nanoscale' dimension becomes important (see Section 24.1.2).

There are, of course, current non-biocidal technologies, notably the commercial fouling-release silicone-based coatings. These are particularly successful for vessels characterised as 'high speed/high activity' but, are less successful for vessels that operate for most of their time at low-moderate cruising speeds, and which spend a significant proportion of their time at anchor, such as navy fleets. There are also issues of durability and the

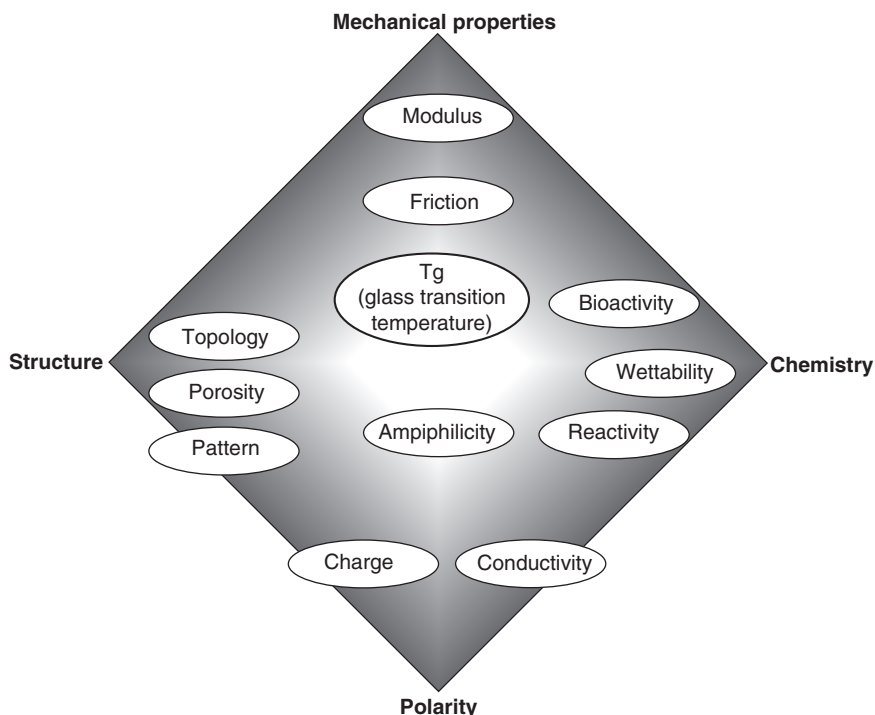
leaching of oils from some coatings. There is therefore considerable scope for either improving the properties of current fouling-release coating designs, or devising alternatives to the silicone-based materials in current commercial use. This is a key objective of the AMBIO project.

24.1.2 Why 'Nano'?

Nanotechnology is science and engineering at the scale of atoms and molecules. It deals with objects and materials that can be made, controlled, or manipulated on the scale of 1 to 100 nanometres. It studies the fundamental principles of these objects and the phenomena and laws that are particular to this length-scale, and is the natural scale of all fundamental life processes.

Biofouling is a prime example of a nanoscale interfacial process. Organisms that foul surfaces in aquatic environments, such as barnacles, algae, bacteria, do so through the secretion of adhesive polymers. The fouling process involves interactions determined within a few nanometres of the surface, the tenacity of adhesion being determined by the molecular interaction of specific structural elements in these adhesives (e.g., amino acid motifs or specific chemical groups) with the substrate, through a range of bonding mechanisms (Smith and Callow 2006). For example, in studies on the influence of topography on attachment of marine bacteria, it has been shown that the almost instantaneous adhesion is influenced by surface imperfections in the nanometre range, i.e. far smaller than the size of bacteria, suggesting that the results were due to a similarity between the size and shape of the imperfections and the size and configuration of the adhesive molecules (Kerr and Cowling 2003). Furthermore, fouling organisms, especially invertebrates and algae, are highly selective in their preferences for certain surfaces. 'Cues' presented by a surface initiate characteristic behaviours which either promote or inhibit attachment. Whilst these have been extensively explored at the micro-scale (e.g., Carman *et al.* 2006; Schumacher *et al.* 2007; Scardino *et al.* 2008) there is relatively little information on the influence of nanoscale features on the settlement of fouling organisms.

Adhesion may thus be regarded as a natural nanoscale process and therefore its ultimate control lies in modifying the properties of surfaces at the nanoscale through novel surface engineering technologies. Figure 24.3 summarises the range of interfacial properties that have been shown by experimentation to be relevant to anti-settlement and/or fouling-release performance of coatings against marine organisms. All of these interfacial properties may be influenced by the nanostructure of a coating, but as yet, few such influences have been systematically investigated. Whilst it is combinations of these parameters that will ultimately determine the overall



24.3 The range of physico-chemical coating properties that can influence settlement and adhesion of fouling organisms (modified after Rosenhahn *et al.* 2008).

performance of a coating for antifouling applications, the systematic manipulation of single interfacial properties to determine which exert the most important influences on fouling organisms is a worthwhile approach. This requires great precision and control over surface properties and in the AMBIO project this is provided by experiments on ‘model’ surfaces in which chemical and physical properties are patterned or tailored at different length scales through self-assembly, nanolithography, photolithography and micro-contact printing.

24.2 Aims of the AMBIO project

The primary aim of the AMBIO project is to explore the potential for innovative, alternative coating solutions using a ‘knowledge-driven’ engineering strategy based on sound correlations between the chemistry of the coating, its physico-chemical and materials properties, and the biological performance. To achieve this ambitious aim requires the integration of a

wide range of scientific and engineering disciplines coupled to state-of-the-art approaches for:

- use of novel materials such as nanoparticles for structuring coatings
- characterising surface properties that are most relevant to biological performance, e.g. environmental AFM for *in situ* analysis of surface topology of fully hydrated coatings
- a biological evaluation programme that facilitates understanding of coatings that succeed or fail at a mechanistic level
- translation of coating concepts from the laboratory scale of the experimental scientist to the level of a formulated coating technology that can be produced and applied on a scale that will allow performance testing in real-world environments.

24.3 Structure of the project

24.3.1 Participants

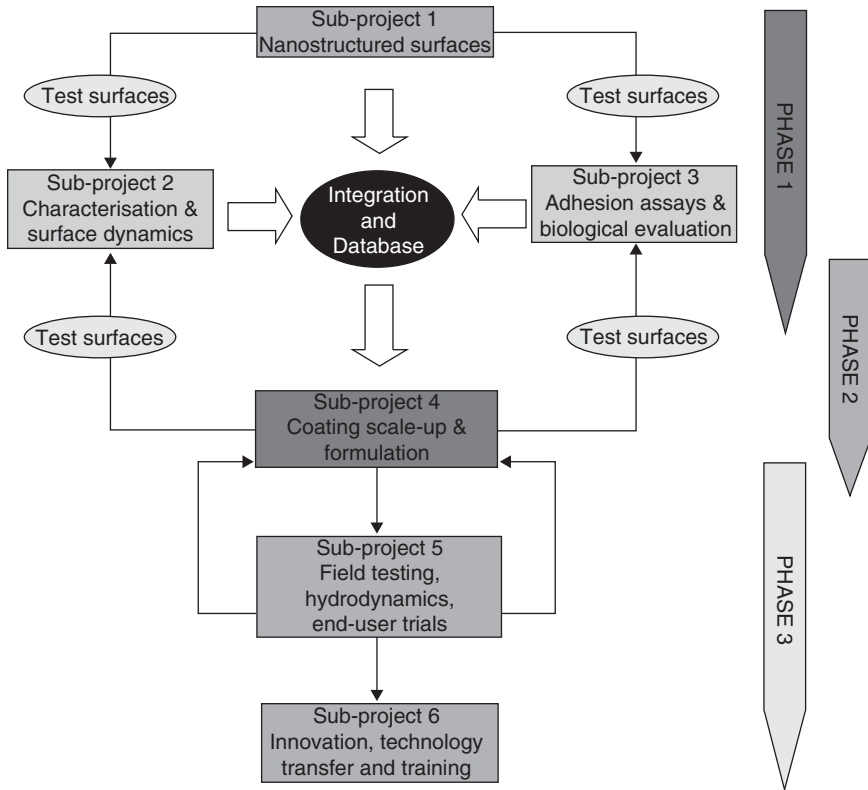
The project contains 31 'Partners' from 14 different nations within the EU, or which have some form of affiliation (Turkey, Norway, Israel). There are 11 universities, 5 research organisations and 15 companies (<http://www.ambio.bham.ac.uk/about/Partners.htm>), the latter including two large multinationals (BASF, Akzo-Nobel (International Paints)) and several SMEs (Fig. 24.2). 'Partners' are members of a consortium with formal contractual obligations to the EC. Intellectual property rights are formalised through a Consortium Agreement.

24.3.2 Work plan

Like all EC-funded projects, AMBIO is structured into a number of work packages with discrete objectives and activities, and these are organised into six sub-projects (Fig. 24.4).

Phase 1 ('Experimental') represents the main focus of activity in the first 3–4 years of the project in which the combination of novel coating and surface-structuring technologies, characterisation of interfacial and materials properties, and biological evaluation was anticipated to generate: a) candidate coatings for further work in the project, and b) new knowledge on the adhesion processes used by fouling organisms.

By the end of the third year it was anticipated that a number of candidate technologies with the appropriate combination of properties would be selected to progress to Phase 2 ('Scale-up and development'), in which experimental coatings will be scaled up and formulated by specialist partners (e.g., Akzo-Nobel) to a point where they are testable in field trials in



24.4 The structure of the AMBIO project.

Phase 3 (years 4–5). For coatings to progress, the following selection criteria were applied:

- good intrinsic antifouling and/or fouling-release properties against representative test organisms
- stable coatings on appropriate substrates in either marine or fresh-water conditions
- the coatings represent conceptual advances in nanoscale engineering for novel coating technologies
- there is a clear end-user application available for field-testing within the project.

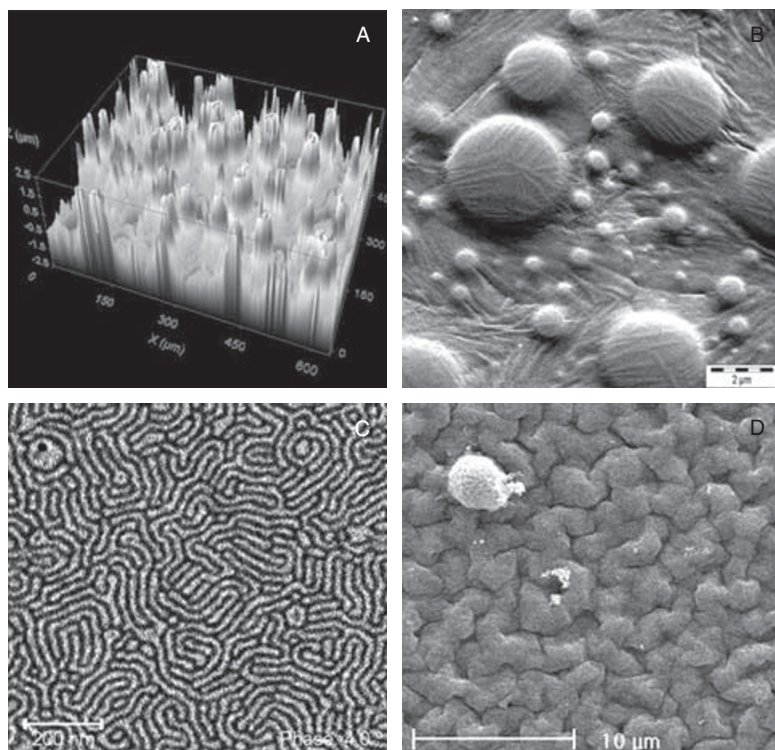
Phase 3 ('Field trials') due to start in March 2008, aims to make quantitative and comparative evaluations of the antifouling efficacy of promising nanostructured technologies through controlled field-scale testing. As a first approach, selected coatings/surfaces will be applied to a range of substrates representative of the range of end-user applications in the project (primed

steel for ship hull coatings, fibre-glass for pleasure craft, polypropylene for fish nets, titanium plates for heat exchangers, membranes for water purification, glass for optical windows, etc.). Replicate substrates will be immersed in suitable environmental conditions for a period of 2–18 months depending on their envisaged end-use. The performance of the technologies will be monitored by assessment alongside industry-accepted standards. For example, in-field adhesion measurements (ASTM D5618) will be used to measure the shear forces required for removal of adult barnacles from fouling release surfaces. Feedback on performance will be provided for further formulation/development work in SP4. The drag characteristics of promising nanostructured technologies will be tested hydrodynamically. Finally a ‘Demonstration Activity’ towards the end of the project will monitor the in-service performance of selected coatings alongside industry-accepted standards for a period of 2–10 months, depending on their envisaged end-use.

24.3.3 Methodological aspects: nanostructured coating designs

Nanostructured surfaces may be classified by their manner of preparation. ‘Bottom-up’ methods use chemical building blocks (molecules, nanoparticles) and self-organisation/supramolecular chemistry, for assembling surfaces with nanosized dimensions. Physical vapour deposition (PVD), chemical vapour deposition (CVD), and electrochemical deposition techniques are used to prepare thin films on top of solid substrates. Coatings of this type are particularly relevant for heat exchanger plates, where it is important that thin coatings are used for efficient thermal transfer, but other application areas for such coatings include optical windows (if they can bond effectively to glass) or water-filtration membranes (providing they do not occlude the pores). Since such methods are analogous to self-assembly of molecules on a surface from solution or the gas phase, it may be considered that these also represent a ‘bottom up’ approach. Bottom-up approaches, especially the polymer-based methods are particularly suited to those applications of AMBIO where a coating has to be applied to structures for aquatic environments, possibly on a very large scale. ‘Top-down’ approaches, e.g. nanolithographic methods, are only feasible for sophisticated, small area applications. Their use in AMBIO is limited to ‘model surfaces’ for exploring adhesion mechanisms.

The major approaches in AMBIO to the creation of novel coatings with tunable nanoscale properties are based on supramolecular chemistry, self-assembly, phase-separation of polymers, superlattices, or molecular building blocks. Appropriate polymers either alone, through spontaneous restructuring or self-assembly, or in combination with nanoparticulate materials, are



24.5 Selected examples of nanostructured coatings in the AMBIO project. A) Nanostructured sol-gel coating containing protruding nanoparticles viewed by confocal microscopy (courtesy TNO). B) SEM of nanostructured surface of solvent-cast polyolefins showing self-assembled surface topologies at micro- and nanoscales (courtesy Gebze Institute of Technology and BASF). C) AFM phase image of a block copolymer coating showing nanoscale phase-segregation of hard and soft blocks (courtesy University of Pisa and BASF). D) SEM image of self-assembled polyelectrolyte surface showing micro- and nanoscale topologies. This surface is highly resistant to the settlement of algal spores; the spherical object (top left) is a rare example of a settled *Ulva* spore (University of Heidelberg and University of Birmingham).

used to create a coating layer presenting a nanostructure to the surface (Fig. 24.5).

In the case of ‘model’ surfaces, the focus at the Universities of Heidelberg and Linköping is on the development of well-characterised tunable surfaces with which to explore the settlement and adhesion properties of fouling organisms. For example ‘top-down’ nanostructuring techniques such as various forms of lithography and self-assembly facilitate independent control of surface chemistry, topography and patterning.

24.3.4 Methodological aspects: evaluation of materials for antifouling technologies

Laboratory-based evaluations of the adhesion of organisms to surfaces have focused mostly on bacteria and have used settlement assays in either static or flow-through systems. Few studies have addressed the settlement and adhesion of higher organisms (algae and invertebrates) and evaluations of non-biocidal antifouling technologies have relied on higher level screening, e.g. raft panels.

A systematic, iterative approach to biological evaluation of surfaces and coatings is used within the AMBIO project. In the experimental phase, the main objective is to test the intrinsic anti-settlement and fouling-release performance of coatings on an experimental scale. Assays are performed on coated microscope slides with representatives of the main groups of fouling organisms, viz. bacteria, algae and barnacles. Coatings are selected for further study based on settlement and adhesion strength data compared to standard surfaces. Epoxy (Epikote) is considered as a negative standard for settlement (attachment) since it is known to foul readily. A silicone elastomer, Silastic®-T2 (Dow Corning) that has good fouling-release properties for higher organisms is the positive fouling-release standard. Assays of model surfaces typically include acid-washed glass as a standard.

Attachment and the adhesion strength of bacterial biofilms is studied at TNO with 3 species of marine bacteria (*Vibrio alginolyticus*, *Cobetia marina* and *Marinobacter hydrocarbonoclasticus*) presented either as a single species or a mixed consortium, and at the University of Dundee with the freshwater bacterium *Pseudomonas fluorescens*. The strength of attachment of the marine bacteria is determined by exposure to hydrodynamic shear provided by a rotor, operating at a speed equivalent to a ship moving at 10 knots. Biofilm is quantified by measurement of fluorescence in a plate reader after staining with a fluorescent dye such as Syto 13. The strength of attachment of *Pseudomonas fluorescens* is determined using a specially constructed apparatus that allows slides to be washed by a dipping process. Cells are quantified using agar plate dilution.

Assays with algae, performed at the University of Birmingham, employ zoospores and sporelings (young plants) of the green seaweed *Ulva* and a diatom, *Navicula*. *Ulva* (syn, *Enteromorpha*) is the most common macroalga that fouls ships and other submerged structures. Dispersal is mainly through motile, quadriflagellate zoospores (approximately 7–8 µm in length), which are released in large numbers and adhere through discharge of a glycoprotein adhesive (Callow and Callow 2006). The number of zoospores that settle on a surface provides information about the antifouling potential of a surface (Callow *et al.* 1997). The settled spores rapidly germinate into sporelings (young plants), which adhere weakly to silicone fouling-

release coatings (Chaudhury *et al.* 2005). The strength of attachment of sporelings is quantified by exposing slides to hydrodynamic forces generated either in a calibrated flow channel (Schultz *et al.* 2000) or from a water jet (Finlay *et al.* 2002). Those coatings that show comparable release of *Ulva* sporelings to that from Silastic® T2 are investigated for the release of hard fouling.

Two assays with the barnacle *Balanus amphitrite* (= *Amphibalanus amphitrite*, Clare and Høeg 2008) are employed at the University of Newcastle. The first provides information about the antifouling properties and quantifies the number of cypris larvae (cyprids) that settle on the test surfaces after 24 and 48 h compared to glass and epoxy (Hellio *et al.* 2004). The second barnacle assay is performed on those surfaces that show promise with the *Ulva* sporeling release assay. Juvenile barnacles are allowed to grow on the test surfaces until sufficiently large to give robust adhesion strength data, typically 12 weeks from settlement of the cyprids. The adhesion strength measurement is performed with a novel custom built apparatus that effectively automates the standard ASTM D5618 method that uses a hand-held force gauge (Conlan *et al.*, 2008). The force gauge automatically pushes the barnacle across the test surface at constant speed and can measure up to 10 N total force. The basal area of each barnacle being tested is calculated using digital scans analysed by image analysis software incorporated into the computer controlled force gauge's software. Adhesive strength per unit area is calculated to give adhesive strength in MPa.

24.3.5 Methodological aspects: digital in-line holography

One of the key ideas for the development of novel anti-fouling coatings is to exploit the 'choosiness' of the settling (adhesion) stages of fouling organisms by creating a deterrent surface. Whilst much can be done through simple assays of the ability of, e.g., motile larvae or spores of algae to settle on a surface, a more comprehensive understanding requires detailed observation of the behaviour of organisms as they explore different types of surface. However, the three-dimensionality of the motion of, for example, a swimming spore of an alga, makes it difficult for conventional optical microscopy to follow the cell throughout its exploration phase from a point far away from the surface to the close vicinity of the surface where sensing occurs. High-speed data acquisition in modern light microscopes allows algal cells to be traced, but the intrinsic two dimensionality of the technique provides only either the projection of the trace (large depth of focus) or the visualisation of one focal plane (small depth of focus). In general, a quantification of motion is difficult with this type of compound microscopy as all 3D coordinates are mandatory in order to perform an accurate calculation of, for example, swimming velocity.

In the AMBIO project a collaboration between the group at the University of Heidelberg and the University of Birmingham has demonstrated the feasibility of an entirely new approach based on digital in-line holography. An outline of the principles behind this approach is presented in Heydt *et al.* (2007) but in brief, the underlying concept is the formation of a diffraction pattern when coherent radiation is penetrating a sample. The undisturbed photons ('primary wavefront' or 'reference wave') and those scattered from the object ('secondary wavefront') interfere and form the observed diffraction pattern. Owing to the presence of the reference wave, it is possible to digitally reconstruct the holograms without phase retrieval, and obtain three dimensional real space information. Heydt *et al.* (2007) were the first to apply this technology to studies of swimming organisms that show surface exploration and settlement/adhesion types of behaviour. They showed that a range of 3D motion patterns could be detected when *Ulva* zoospores explore a surface, demonstrating the potential of digital in-line holography to image and quantify exploratory patterns of such organisms close to surfaces. More recently, Heydt *et al.* (in preparation) have shown that quite different patterns of exploration can be detected when zoospores explore surfaces on which they 'prefer' to settle (such as a hydrophobic fluorinated surface) compared with less congenial hydrophilic surfaces such as those presented by polyethylene glycols. An extension of this type of analysis to surfaces showing various types of surface topology is underway.

24.4 Selected emerging results

The project aims, as stated above, can be translated into 4 key questions:

- 1 Which nanodesigns show good antifouling properties against a range of test organisms?
- 2 How does nanostructure influence adhesion of organisms?
- 3 How do coatings respond to the aquatic environment and how do nano-fillers influence this?
- 4 Which design concepts show most promise as durable antifouling coatings for a range of 'real-world' applications?

At the time of the preparation of this article the project has just started its 4th year and for the first three questions notable progress has been made. However, most of the scientific results have yet to find their way into the primary literature. Furthermore, some results contain significant intellectual property and therefore cannot yet be disclosed. Therefore, in this section it is only appropriate to give a broad overview of emerging results suitable for the public domain.

24.4.1 'Practical' coatings

At the end of year 3 a total of 15 different coating types was selected for scale-up and formulation (and 3 provisional patents were filed). The details of the selected coatings cannot be presented here but, at a more general level significant advances made so far include the following:

- Silicone elastomers have been reinforced through the incorporation of multiwall carbon nanotubes (MWCNTs) and sepiolite nanoclays, (collaboration between the group of Philippe Dubois at the University of Mons-Hainault and Nanocyl, a specialist producer of carbon nanotubes). These nanocomposites showed enhanced fouling-release performance in laboratory assays with algae and barnacles compared with unfilled controls (Begbeider *et al.* 2008a) as well as in short-term field tests. The use of MWCNTs is particularly significant since the improved performance is obtained at low loadings of the nanofiller (0.05–0.2% by weight). At this level of loading, the bulk properties of the coatings (tensile moduli and cross-link density) appear to be unchanged but the coatings become slightly more hydrophobic. A significant change in shear-thinning behaviour has been detected (Begbeider *et al.* 2008b), which is particularly significant in terms of ease of application of such coatings to surfaces. The rheological properties have been attributed, on both theoretical and experimental grounds, to a strong molecular affinity between the siloxane chains and the MWCNTs, via CH- π interactions involving methyl groups of the PDMS and aromatic rings of the MWCNTs (Begbeider *et al.* 2008b). This affinity is also important in reducing the likelihood of MWCNTs being released into the marine (or any other) environment, which is relevant to the current debate on potential issues of nanoparticle toxicity. Current studies at the University of Mons-Hainault aim to understand how these molecular interactions generate the enhanced fouling-release performance with a focus on changes in surface topological properties.
- The University of Pisa has designed coatings based on fluorinated and siloxane-based block co-polymers, which through self-assembly and phase segregation generate a nanostructuring effect that influences surface energy and topology. These coatings show good fouling-release performance in laboratory tests with macroalgae and adult barnacles (Universities of Birmingham and Newcastle).
- Sustech and Linköping University have designed coatings based on PEGylated and other hydrogel-type materials. These have some history of application as antifouling surfaces in biomedicine, but have not been systematically investigated for marine antifouling. Antifouling tests show some effectiveness against marine and freshwater bacterial biofilms

(TNO and University of Dundee) and the settlement and adhesion of green algal spores, diatoms and barnacle cyprids (Universities of Birmingham and Newcastle) (Ekblad *et al.*, 2008). Stability of hydrogels in seawater is an issue but field-application to end-windows of optical instruments will be explored, since for such applications an effective coating that lasts only a few months is still useful.

- Gebze Institute of Technology has designed superhydrophobic, polyolefin-based coatings and blends which produce surface topologies that can be adjusted by the deposition techniques used. These coatings are effective against soft fouling and have some potential for use in fresh-water environments.
- TNO has produced nanocomposite coatings based on sol-gel chemistry, in which the length scale of surface topology can be controlled by the chemistry of the cure reaction and the chemical functionalisation of the nanoparticles. These are effective against some of the fouling organisms tested.
- Good progress has also been made in developing bioactive model coatings using enzymes. The Institut für Polymerforschung, Dresden, has successfully explored strategies for the use of immobilised proteolytic enzymes to prevent marine biofouling by hydrolysing the adhesive proteins secreted by the fouling organisms. Subtilisin A, a proteolytic enzyme used in washing-powders, was randomly immobilised by covalent attachment to different maleic anhydride copolymer coatings. The enzyme-containing coatings reduce the settlement and the adhesion strength of spores of *Ulva linza* and the adhesion strength of diatom cells.
- Ultra-thin, low surface energy metal-polymer nanocomposites have been produced through chemical and physical vapour deposition techniques (TEER Ltd and University of Dundee) or electro-plating (CIDETC). These, typically low-energy surfaces, show efficacy in reducing bacterial biofilm formation and adhesion (e.g., Zhao *et al.* 2007) and in some cases show good release of macrofouling algae, which is unexpected for a non-elastomeric coating (Akeso *et al.* 2009). These surfaces are now being evaluated for use on heat exchanger tubes and plates where only thin coatings can be used in order to minimise the effect on heat transfer.

24.4.2 Model surfaces

Other fundamental advances in the project follow from the use of controlled, well-characterised model nanostructured surfaces (prepared, for example, by 'soft' lithography and self-assembly) that vary systematically in properties such as surface energy, charge density, friction and topology. Two good examples in AMBIO of the influence of molecular, or nanoscale

properties on the adhesion of fouling organisms are reflected in recently published work (Bowen *et al.* 2006; Schilp *et al.* 2007).

It is well known that protein resistance of self-assembled monolayers (SAMs) of hexa(ethylene glycols) (EG₆) is dependent on the alkoxyl end-group termination of the SAMs, which in turn determines wettability (Herrwerth *et al.* 2003). Schilp *et al.* (2007) used the same series of hexa(ethylene glycols) to examine the correlation between protein resistance and the settlement and adhesion of zoospores of the macroalga *Ulva* and cells of the diatom *Navicula*, which adhere to the substratum through the secretion of protein-containing glues. Results showed that adsorption of fibrinogen, and attachment of both *Ulva* zoospores and cells of the diatom *Navicula* follow the same general trend and showed a general dependence of the wettability of the surfaces, i.e. the number of attached cells increased as wettability (hydrophobicity) increased. However, on closer inspection it was shown that the most hydrophilic SAM, EG₆OH, and to a lesser extent EG₆OMe, actually promoted settlement of *Ulva* zoospores but the settled cells were so weakly adhered that they floated off the surface as 'rafts' following even the slightest disturbance of the sample. The low numbers that adhere to this surface presumably reflects the fact that the secreted glycoprotein adhesive is unable to bond to the surface because water cannot be displaced from the interface, a condition which is essential to bring the bioadhesive molecules in direct contact with the SAM surface. The hydration of the SAM and adsorption of hydroxide ions to its end-group declines with increasing alkyl chain length and appears to moderate protein adsorption as well as the adhesion of higher organisms such as zoospores of *Ulva* and cells of *Navicula*.

In another example of the use of SAMs to provide a systematic variation in surface properties, Bowen *et al.* (2006) showed that frictional resistance of a surface is important in regulating detachment of *Ulva* spores and diatom cells under shear stress. A series of alkanethiol SAMs of different chain length (C8–C18) was prepared. These have very similar wettabilities but vary in their frictional properties. Results show that the adhesion strength of diatom cells and algal spores to the SAMs decreased with increasing alkyl chain length at C14 and above. This is the point of transition between the lower chain length, amorphous, less-ordered layers with high frictional coefficients, and the higher chain lengths which produce more crystalline, more lubricious structures less able to dissipate energy throughout the SAM, resulting in greater adhesive bond failure.

24.5 Acknowledgements

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